# FORCE

## Grade 7 - Semester 1

Force is a push or a pull action on an object resulting from an object's interaction with another object. The action can cause acceleration, change in velocity or alter its shape. To measure force, we measure it in newtons (N).

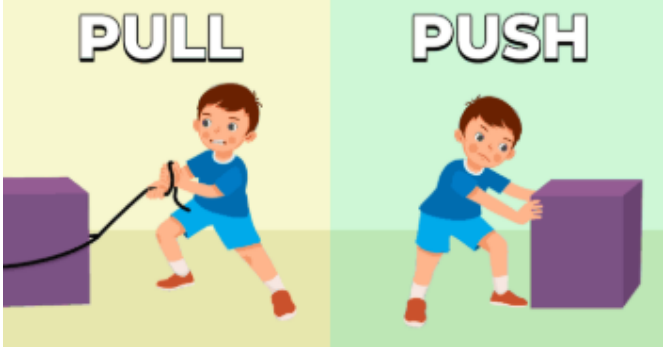


Figure 1, Push and Pull

**Newton's Law**

In force, we use Newton's law which was created by Isaac Newton. Newton's Law was found when Sir Isaac Newton noticed that a force was pulling down the apple from the tree, making the apple fall down. There are 3 laws which are, Newton's 1st law, Newton's 2nd law and Newton's 3rd law.

Newton's first law explains that an object at rest will stay at rest and an object in motion will stay in motion unless acted upon by an unbalanced force.

Newton's 2nd law tells us that an acceleration depends on the force applied and the mass of the object

Lastly, Newton's 3rd law states that for every action, there is an equal and opposite reaction.

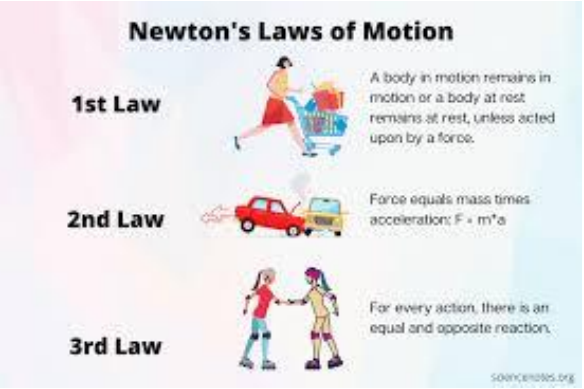


Figure 2 Newton's Laws of Motion

**FORCE DIAGARAM**

Force diagram is a tool used in physics to visualize all forces acting on an object, it's usually represented by a box or a dot. A force diagram has 2 types of forces which are contact and action-at-a-distance force. Contact force is any force that acts on an object that is physically touching one another (sledding : frictional force). While action-at-a-distance force is a force that happens without needing any physical contact.

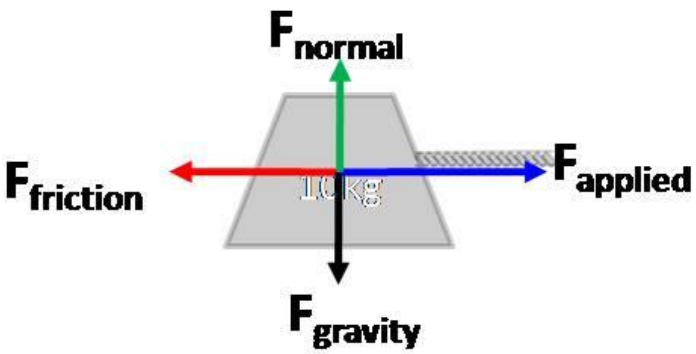


Figure 3 Example of Force Diagram

If we break it down, there are many forces in contact forces. In contact forces there are :

1. Frictional force → A soccer ball slowing down
2. Tension Force → Pulling a rope in tug of war
3. Normal Force → A lamp placed on a table
4. Air resistance force → A parachute opening, slowing the skydiver
5. Applied force → A person pushing a box

6. Spring force → A pogo stick

Figure 4 Examples of Contact Forces

There are also many forces in action-at-a-distance force which are :

1. Gravitational Force → Apple falling down from a tree
2. Electrical Force → Balloon sticking to the wall after being rubbed on hair
3. Magnetic Force → Magnet sticking to a refrigerator

Figure 5 Example of Non Contact Force (Action-at-a-distance force)

**Formula** :

In force, there is a formula which is F = ma. F stands for force, m for mass and lastly a represents acceleration of an object.

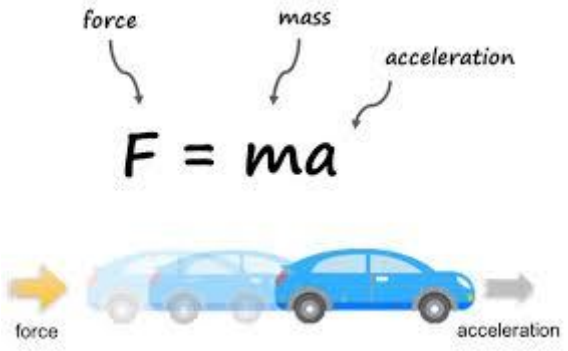


Figure 6 F = ma Formula Example